

ENERGY AI

HACKATHON 2026

Zewail City of Science and Technology

Powered by Solship · Two-Day Competition · 30 Teams

Registration Form



ABOUT SOLSHIP



Turning Sustainability into Profitability

Solship is an AI-driven energy management platform that helps businesses with solar PV and battery storage automatically decide when to store, buy, or sell electricity — maximising returns from every unit of energy generated.



2×

Revenue Increase

30%

Cost Savings

4 yrs

Payback Period

5

Pilot Customers*Residential & Commercial/Industrial sites in Italy*

1 MW

Total Installations*PV + Battery capacity across active sites***R****Residential***6–7 kWp PV · 10–20 kWh battery
Milan area & Parma, Italy***C****Commercial & Industrial***40–743 kW PV · 44–130 kWh battery
Mottolini, Autotorino, Fiera Bolzano***AI****AI Optimisation***Rolling-horizon MPC deployed
on every active site*

HOW SOLSHIP WORKS



01

PREDICT

Load Forecasting

Deep learning models ingest historical consumption, solar generation, temperature, and price signals to forecast load at 15-minute resolution.

02

OPTIMISE

Battery Dispatch

A rolling-horizon Model Predictive Controller solves the optimal charge/discharge schedule every step — buying cheap, selling expensive.

03

EARN

Maximised Savings

The system automatically arbitrages time-of-use tariffs, exports surplus at peak prices, and supports grid emergencies for additional revenue.

AI sends a signal → Hardware communicates with all inverters → Instant Response



THE CHALLENGE

Load Forecasting · Battery Dispatch · Rolling MPC

01

A real-world energy optimisation problem from an Italian residential site

PROBLEM OVERVIEW



Your Mission

You are given a residential site in Italy connected to the utility grid, a 9 kWp rooftop solar PV system, and a 16 kWh lithium battery.

Build two things:

 **A load forecasting model**
trained on 2024 data, applied to 2025

 **A rolling-horizon battery**
dispatch controller that cuts the electricity bill

Power Sign Convention

Variable / Sign	Meaning
$P_{\text{battery}} < 0$	Battery charging
$P_{\text{battery}} > 0$	Battery discharging
$P_{\text{grid}} > 0$	Importing from grid
$P_{\text{grid}} < 0$	Exporting to grid

SITE LAYOUT

Solar PV

9 kWp

Grid

6 kW limit

Battery

16 kWh / 8 kW

Load

Must be served 100%

The site load must be fully served at every 15-minute interval without exception

SYSTEM SPECIFICATIONS



9 kWp

Solar PV

Peak at STC

16 kWh

Battery Capacity

Usable energy

8 kW

Max Charge/Discharge

Symmetric limit

90%

Round-trip Efficiency

v0.90 per direction

6 kW

Grid Connection

Import & export limit

50%

Initial SoC

Start of 2025 period

Italian Residential Time-of-Use Tariff

Band	Buy Price	Active Hours
F1 — Peak	€ 0.2540/kWh	Mon–Fri 08:00–19:00 (excl. holidays)
F2 — Shoulder	€ 0.2682/kWh	Mon–Fri 07–08 & 19–23 Sat 07:00–23:00
F3 — Off-peak	€ 0.2440/kWh	Nights, Sundays & national holidays

Sell price at each timestep is provided in the dataset · $\Delta t = 0.25$ h · All power in kW, energy in kWh

DATASET & RULES



Excel file — two sheets: 2024 (training) and 2025 (test)

Column	Unit	Description
timestamp	datetime	15-min interval start, Italian local time (CET/CEST)
load_kw	kW	Site electrical load (always ≥ 0)
pv_kw	kW	PV generation (zero at night)
buy_price	€/kWh	Grid import price (F1/F2/F3 band)
sell_price	€/kWh	Grid export price (from market data)
p_battery_kw	kW	Actual battery power recorded on site → Baseline A signal

TRAIN 2024 data

*Build & tune your
forecasting model*

TEST 2025 data

*Report all metrics &
compute all bills here for 2 specified months*

⚠ Data integrity: A multi-week window in the 2025 p_battery_kw column contains an energy-balance anomaly. Reconstruct SoC from scratch before optimising.

LOAD FORECASTING



Train on 2024 data · Forecast April and September 2025 at 15-min resolution · Moving Forecaster - Any algorithm

RMSE

kW

Root Mean Squared Error
Penalises large errors more

Required

MAE/MAPE

kW

Mean Absolute Percentage Error
Robust to outliers

Required

NRMSE

%

$\text{RMSE} \div \text{mean}(\text{load}) \times 100$
Primary ranking metric

★ Ranking

Practical Tips

- Hour-of-day, day-of-week, month, and holiday flags are effective calendar features
- Temperature data (sourced externally) can improve thermally-sensitive load models
- Validate on an internal hold-out before ever touching 2025 — no data leakage allowed
- Report NRMSE on the Day 2 surprise dataset without retraining — tests generalisation

NRMSE is the primary ranking metric — it is normalised, so comparable across both sites

⚠ ROLLING-HORIZON MPC IS REQUIRED

A one-shot batch optimisation over the full 2025 year is not acceptable — it violates causality and produces artificially inflated savings.

Objective Function — Minimise Total Electricity Bill:

$$Bill = \sum [P_{grid}(t) \times buy_price(t) \times \mathbb{I}(import) - |P_{grid}(t)| \times sell_price(t) \times \mathbb{I}(export)] \times \Delta t$$

Hard Constraints

Grid Power

$$-6 \text{ kW} \leq P_{grid}(t) \leq 6 \text{ kW}$$

Battery Power

$$-8 \text{ kW} \leq P_{battery}(t) \leq 8 \text{ kW}$$

State of Charge

$$0 \leq SoC(t) \leq 1$$

CORE TASK — Required for all teams

- 1 Train a load rolling forecasting model on the 2024 data
- 2 Apply it to generate 2 months in 2025 load forecast
- 3 Build a rolling-horizon battery dispatch controller using your forecast
- 4 Compare your controller's 2025 electricity bill against both baselines
- 5 Report RMSE, MAE/MAPE, and NRMSE on the 2025 test set
- 6 Oracle gap: re-run with actual 2025 load as input — report savings difference in € and %

EXTENSION 1 — Optional +5 bonus pts

Horizon sensitivity: run the 2025 optimisation for ≥ 3 different horizon lengths H (e.g., 1h / 6h / 24h). Report savings vs. computation time table. Justify your chosen H .

SCORING & PRIZES



Controller savings vs. Baseline A

% savings vs. historical operation on 2025

35 pts

1st

20,000 EGP

Forecasting accuracy — NRMSE on 2025

Must report RMSE, MAE, and NRMSE

25 pts

2nd

10,000 EGP

Generalisation — NRMSE on surprise dataset

NRMSE on Day 2 dataset vs. 2025 baseline

25 pts

3rd

5,000 EGP

Reasoning & presentation clarity

Quality of design justifications

15 pts

+ 5 bonus pts for Extension 1 (horizon sensitivity)

Tiebreaker: (1) controller savings (2) NRMSE surprise dataset (3) NRMSE 2025 test set

EVENT SCHEDULE



DAY 1 — Explore · Forecast · Formulate

08:30 Registration

09:00 Welcome Remarks — Dr. Ahmed Abdelsamea

09:15 Problem Statement Walkthrough

09:45 Dataset Distribution & Environment Check

10:00 Hacking Block 1 — EDA & Load Forecasting

13:00 Lunch & Mentor Office Hours

14:00 Hacking Block 2 — Forecasting & Optimizer

17:00 Closing Remarks & Day 1 Submission

DAY 2 — Optimise · Generalise · Present

09:00 Welcome Remarks — Dr. Ahmed Abdelsamea

09:30 Hacking Block 3 — Rolling-Horizon Controller

13:00 Lunch & Final Mentor Session

14:00 Hacking Block 4 + Surprise Dataset Released

15:00 Final Submission Deadline

15:15 Team Presentations (30 teams × 3 min)

17:00 Judging Deliberation

17:30 Prize Announcement & Closing Ceremony



Ready to Optimise?

*Join 30 teams competing to build the best
AI-powered battery dispatch controller*

WHEN

Two-Day Event

WHERE

Zewail City, Egypt

TEAMS

Up to 2 per team

PRIZES

35,000 EGP total